



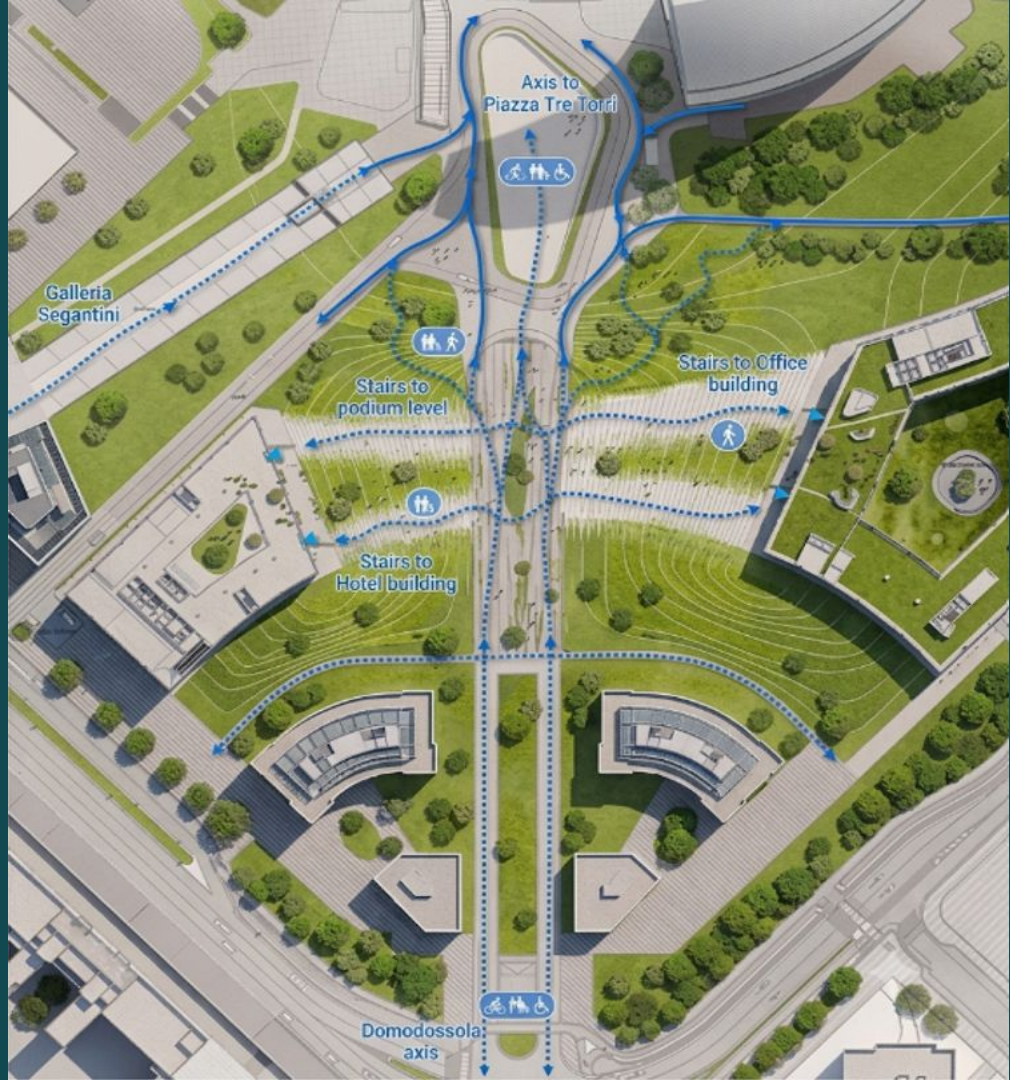
# AccessMap

*Can data predict which sidewalks are accessible?*

Group 13A · AI4ALL Ignite Summer 2026

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Live demo: [accessmap-13a.streamlit.app](https://accessmap-13a.streamlit.app)



# The Problem

Wheelchair users can't tell if a sidewalk is passable before they get there.

Too narrow, curb too high, they find out on the spot. A wrong route isn't an inconvenience; it can end a trip.

Maps show that a road exists. They don't show whether a wheelchair can use it.

**Research question:** Can data predict which sidewalks are accessible for wheelchair users?

## Why it matters



Wheelchair users, caregivers, seniors plan routes with confidence



Cities collect sidewalk data but rarely turn it into anything usable



A working model plugs into any routing app as a scoring layer

# Two Datasets, One Question



PMR — Amsterdam sidewalks

72,274

measured segments

Physical measurements: width, curb height, crossings. Measures the physical world.



Housing — US addresses

10,104

crowd-labeled rows

Crowd-sourced “sidewalk ok?” labels judged from Google Maps street images.

**No geographic overlap.** One is meters in Amsterdam, the other spans a continent in degrees. Same pipeline runs on each; results are compared at the model level. Rows are never merged.

# The Algorithms

## Random Forest (main model)

**Type:** Supervised classification many decision trees vote

**Inputs:** PMR: width, curb height, crossing, length. Housing: house type + crowd-confidence scores

**Output:** accessible / not accessible, with a probability

### Why we chose it (pros)

- Handles mixed features well
- Feature importance is readable — that's how we caught the bias
- Strong baseline without heavy tuning

### Trade-offs (cons)

- Can't invent signal that isn't in the features
- Happily rides a biased shortcut feature
- Less interpretable than a single tree

## Supporting tools

### ModernBERT

Embeds each address into a 768-dim vector (6,425 addresses)

### FAISS

Similarity search over those vectors: “find addresses like this one,” ranked by distance



# How the Project Evolved

**V1**

## Baselines

Both models trained. Two problems surface: Housing below baseline, PMR over-flags.

**V2**

## Bias fix

Drop state from Housing, retrain. Result: a null finding — and an honest one.

**V3**

## Precision fix

PMR threshold tuning fails. Root cause found in features; curb height added.

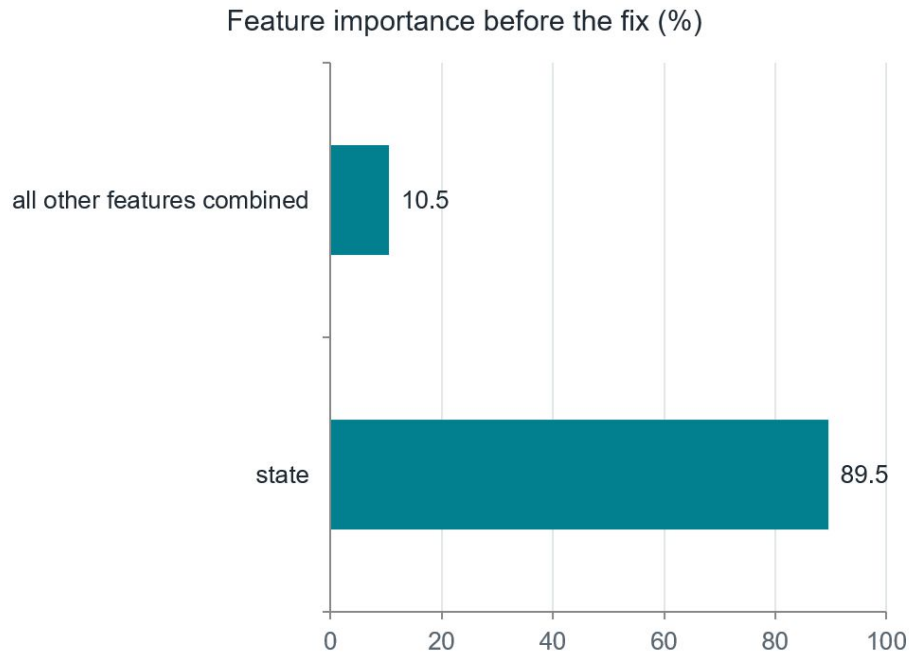
**V4**

## Deployment

Streamlit app built, tested locally + on the cloud, live for anyone to use.

*We didn't get it right the first time — and that's the story worth telling.*

# Challenge 1 — The Housing Model Learned Geography

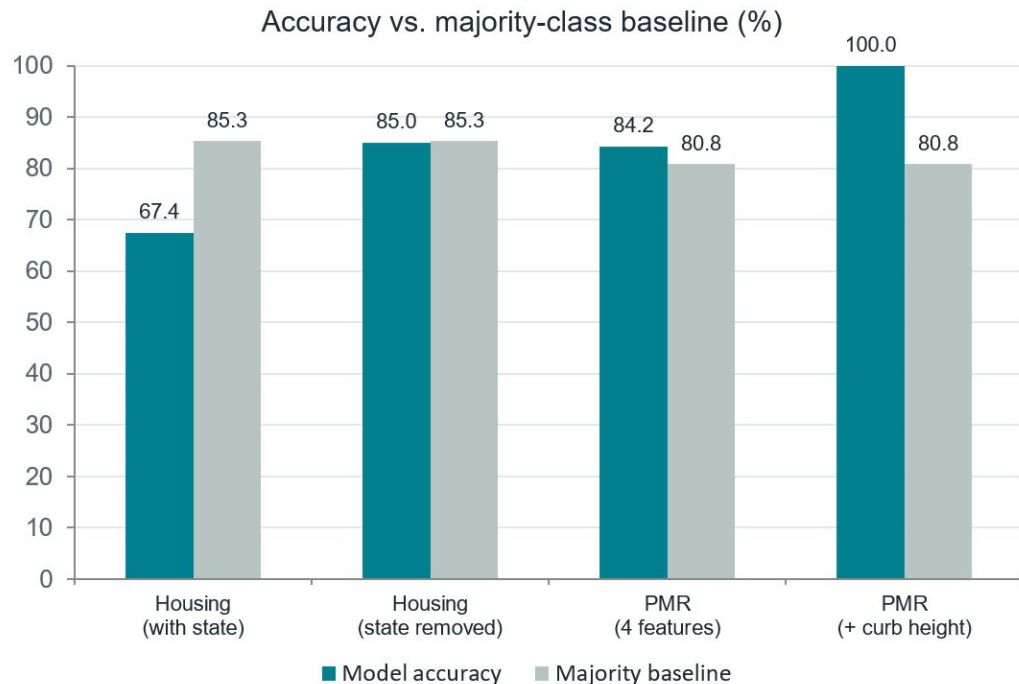


67.4% accuracy vs. an 85.3% baseline — worse than guessing.

One feature, **state**, drove 89% of every decision. The model wasn't judging house condition. It was memorizing which states have more “bad sidewalk” labels.

**This is a bias problem, not a tuning problem.**

# The Fix Revealed the Real Finding



**Housing:** 85.0% ties the 85.3% baseline. Recall on bad sidewalks: 0.01. Without state, no signal is left — and the raw data has no other features to add.

**A null result is a result.** “This dataset can't predict sidewalk condition, and here's why” is our headline bias finding — not a failure to hide.

# Challenge 2 — Confidently Wrong, Not Unsure



## Attempt 1: move the threshold — failed

PMR over-flagged “not accessible” (precision 0.55). We raised the cutoff from 0.5 to 0.65. Nothing changed.

14 / 14,455

test predictions sat anywhere near the cutoff. Probabilities were pinned at 0 or 1 — the model wasn't hesitating, it was confidently wrong. A threshold can't fix confident mistakes.



## Attempt 2: fix the features — worked

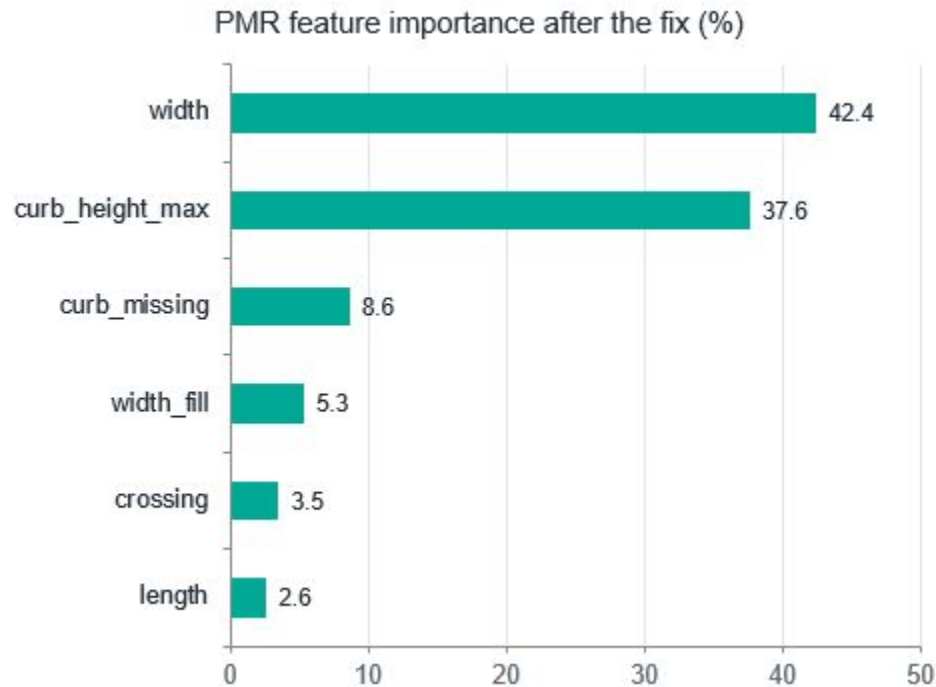
Our label rule = width + curb height. But the model only ever saw width — it was guessing the curb half blind.

We added curb\_height\_max and a missing-value flag (74% of rows have no curb measurement — the missingness itself is information).

*Lesson: check the probability distribution before reaching for a threshold.*



# Evaluation — Read the Numbers Honestly



## **PMR: 100% accuracy.**

Expected, not impressive. The two label inputs (width + curb height) now top the chart — the model recovers the rule we wrote. It proves the pipeline is consistent, nothing more.

## **Housing: 85.0%.**

That's the majority-class rate, not skill. The honest conclusion stands: house-level features alone can't predict sidewalk condition.

# Live Demo

[accessmap-13a.streamlit.app](https://accessmap-13a.streamlit.app)



## Sidewalk Checker

Enter width, curb height, crossing, length →  
accessible / not accessible + confidence



## Address Search

Pick an address → FAISS returns similar  
addresses with sidewalk labels + distances



## About & Limits

What the app does, what it refuses to do, and  
why — honesty as a feature

**One deployment decision we're proud of:** the app does NOT predict sidewalk condition for US addresses. We tested that model — it has no signal — so the app reports the real crowd label instead of pretending to predict.

# Results, In Plain Language

## What were the results?

PMR (Amsterdam): 100% accuracy after adding curb height. Housing (US): 85.0% vs. an 85.3% majority baseline once state was removed — recall on bad sidewalks just 0.01. Address search: FAISS retrieval works and ships in the live app.

## What do the results mean?

The PMR score means the model consistently recovers the width + curb rule we wrote — proof the pipeline works, not superhuman skill. The Housing tie with the baseline is an honest null result: house-level features alone can't predict sidewalk condition.

## How did we get them?

The same recipe on both datasets: clean, encode, train a Random Forest, score on a held-out 20% test set, then audit feature importance. That audit — not the accuracy score — is what exposed the state bias and the missing curb feature.

## How accurate are they?

PMR: 100% on 14,455 held-out segments — but scored against our own rule-based label; only 50 verified labels exist, too few to test on. Housing: no better than always guessing “ok.” We trust the audited pipeline more than any single number.

## Deployment

# Why is the app so simple — and how reliable is it?

### Why we keep the app simple...

Every feature maps to a result we validated. We ship the two things that work — the PMR checker and the address search — and we refuse to ship US sidewalk prediction, because we tested that model and it has no signal. Fewer moving parts also means fewer failure points for the people who rely on the answer.

### How reliable is it...

Confidence now comes with an explanation: it's the share of the forest's trees voting for that label, shown with the factors that drove the call. Inputs are range-checked, missing curb heights use the same fill value as training, and every known limit is stated inside the app itself.

# Impact — Both Sides



## Positive

- Wheelchair users and caregivers plan routes before leaving home
- Cities can find clusters of bad segments and prioritize repairs
- Accessibility becomes a data layer any routing app can adopt



## Negative / risks

- A biased model could systematically mislabel whole regions — exactly what our Housing model did before the fix
- False “accessible” labels are worse than no label: someone plans a trip they can't finish
- Rule-based labels (ours) can encode wrong assumptions if never validated against real users



## Essential Question

# How can the process of creating AI/ML solutions amplify or mitigate bias in our case?

### It amplified bias when...

we let a geographic proxy (state) dominate training. The model looked like it was scoring house condition; it was really scoring ZIP codes. High feature importance made an unfair shortcut look like intelligence.

### It mitigated bias when...

we audited feature importance, removed the proxy, retested, and reported the honest null result instead of a prettier number. The same ML process that created the bias also gave us the tools to catch it — if we chose to look.

# Next Steps



## Rule-free PMR experiment

Retrain on only features outside the label rule (crossing, length, width\_fill): can accessibility be guessed without measuring? The one test where the model can show real skill.

Week 13



## Validate the label

Compare our width/curb rule against the 50 verified accessibility labels; report agreement.

Week 13



## Location-aware retrieval

Add coordinates to the PMR index so retrieval ranks within tied groups instead of returning exact ties.

Week 14



## Individual roadmaps

Each member continues per their My AI Roadmap; repo stays public as a portfolio piece.

Ongoing

# Citations & Sources

## Data sources

- City of Amsterdam Open Data — PMR sidewalk accessibility dataset (72,274 segments)
- CrowdFlower / Figure Eight “Data for Everyone” — Housing: Wheelchairs dataset (10,104 crowd-labeled US addresses)

## Related work & methods

- Breiman, L. (2001). Random Forests. Machine Learning, 45(1).
- Johnson, J., Douze, M., & Jégou, H. (2019). Billion-scale similarity search with GPUs. IEEE Transactions on Big Data. (FAISS)
- Warner, B., et al. (2024). Smarter, Better, Faster, Longer: A Modern Bidirectional Encoder. (ModernBERT)
- Mehrabi, N., et al. (2021). A Survey on Bias and Fairness in Machine Learning. ACM Computing Surveys.
- Saha, M., et al. (2019). Project Sidewalk: A Web-based Crowdsourcing Tool for Collecting Sidewalk Accessibility Data at Scale. ACM CHI.

Live app: [accessmap-13a.streamlit.app](https://accessmap-13a.streamlit.app)    Code + docs: [github.com/zhang-systems/ai4all-accessmap](https://github.com/zhang-systems/ai4all-accessmap)



# Thank You!

*Questions? Try the app yourself while we talk.*

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